Three local interactions of the ten-dimensional self-dual five-form that no stress-tensor flow generates

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Non-linear theories of a chiral four-form gauge field in ten dimensions are usually constructed by deforming a seed theory along a flow built from stress-tensor invariants, as in lower dimensions where such flows generate everything available. Whether that construction reaches the whole space of local interactions in ten dimensions has not been decidable, because the space itself was not known at any fixed order. We determine it exactly at quartic-squared order: the space of local invariants of degree ten in the self-dual five-form is 14-dimensional, the subspace reachable by the stress-tensor flow is 11-dimensional, and the quotient is exactly 3-dimensional. We give explicit representatives of the three unreachable directions, no proper subset of which spans the quotient. Products of lower-degree invariants are all reachable, so the missing directions are not built from anything simpler. The result is obtained in exact arithmetic and recovered independently in graph, block-tensor and spinor descriptions related by a certified equivariant map. Any deformation scheme intended to cover ten-dimensional chiral four-form dynamics at this order must therefore enlarge its generator set by at least three independent structures.

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Introduction. A chiral p-form gauge field carries a self-dual field strength, and in ten dimensions that is the self-dual five-form F of type IIB supergravity. Writing a non-linear theory of such a field is delicate: the self-duality constraint is not a Lagrangian equation of motion, and the standard route to interacting theories is deformation — start from the free theory and flow along a vector field built from invariants of the stress tensor, the higher-dimensional analogue of the $T\bar{T}$ construction [1].

In four and six dimensions this is not merely a convenient method, it is essentially the only one needed: the space of available local invariants is small enough that the flow reaches it. Ten dimensions is structurally different. The self-dual five-form has 126 components and the Lorentz group has dimension 45, so a generic configuration has trivial stabiliser and there are 81 functionally independent invariants [1]. That the flow cannot be universal in this setting is known. What has not been available is any statement of *how* it fails — by how much, in which directions, and whether the shortfall is generated by simpler structures.

The obstacle was that the space of invariants was not known at any fixed order. Twelve explicit degree-ten structures appear in the literature [2], given as index expressions, not claimed to be complete, and with their linear relations undetermined. Without the ambient space one cannot ask what a construction misses.

We remove that obstacle at degree ten and answer the question exactly.

Setup. Work in signature $(-,+,\dots,+)$ with $\epsilon_{01\dots9} = +1$. For a five-form in ten dimensions $\star^2 = (-1)^{p(d-p)}\,\mathrm{sgn}\det\eta = +1$, so real self-dual forms exist; $\mathcal{F}$ denotes the 126-dimensional space of them.

By a *local invariant of degree* n we mean a Lorentz-invariant polynomial of degree n in the components of F, with no derivatives. These are the scalars available to a local deformation at that order. Write

- $\mathcal{A}_{10}$ — all degree-ten invariants;
- $\mathcal{P}_{10} \subset \mathcal{A}_{10}$ — the span of products of lower-degree invariants;
- $\mathcal{D}_{10} \subset \mathcal{A}_{10}$ — the span of degree-ten invariants generated by the stress-tensor flow from the seed theory;
- $\mathcal{Q}_{10} = \mathcal{A}_{10}/\mathcal{D}_{10}$ — the obstruction quotient.

$\mathcal{D}_{10}$ is defined by the flow of Ref. [1], generated from the stress tensor of the free theory; its precise definition, and every assumption below, is stated in the End Matter.

Result. **Theorem.** *Under the hypotheses stated in the End Matter,*

$$\dim\mathcal{A}_{10} = 14, \qquad \dim\mathcal{D}_{10} = 11, \qquad \dim\mathcal{Q}_{10} = 3, \quad (1)$$

and $\mathcal{P}_{10} \subset \mathcal{D}_{10}$. There exist three degree-ten invariants whose classes span $\mathcal{Q}_{10}$; within that fixed set, removing any one leaves only 2 independent classes, so no proper subset spans the quotient.

Three consequences are worth separating.

The deficit is three, with explicit representatives. Not "at least one", which is what non-universality alone gives, but a definite dimension with generators exhibited in the

End Matter — and a rational one. $\mathcal{D}_{10}$ is the only dimension here assembled by admitting directions that raise the rank over $\mathbb{F}_p$, which would leave the deficit an upper bound. Re-running that closure in exact rational arithmetic, after lifting the flow targets by CRT and rational reconstruction validated against a held-out prime, gives $\dim_{\mathbb{Q}} \mathcal{D}_{10} = 11$ and hence $\dim_{\mathbb{Q}} \mathcal{Q}_{10} = 3$, with an explicit non-vanishing integer minor as witness.

The missing directions are irreducible. Because $\mathcal{P}_{10} \subset \mathcal{D}_{10}$, every product of lower-degree invariants is reachable. The three directions are therefore not obtainable by composing lower-order results, and no iteration of a lower-degree construction produces them.

The count is representation-independent. $\mathcal{Q}_{10}$ is computed in three descriptions — contraction graphs, block tensors built from $M_\mu{}^\nu = F_{\mu abcd}F^{\nu abcd}$ and $N_{abc}{}^{def} = F_{abclm}F^{deflm}$, and gamma-traceless symmetric bispinors — related by a certified equivariant isomorphism, and all three give 3.

Method and verification. Everything is computed in exact arithmetic over finite fields rather than in floating point. The reason is not fastidiousness: invariant values span an enormous dynamic range at degree ten, so a floating-point rank requires choosing a tolerance, and that choice is not innocuous. Two runs of the same finite-difference calculation at different sample points reported ranks of 35 and 81; the discrepancy is entirely an artefact of normalising rows that had decayed to rounding noise. Over $\mathbb{F}_p$, zero means zero.

Invariants are represented as contraction graphs, canonicalised so that two labellings of the same contraction are identified, and evaluated at self-dual five-forms sampled over $\mathbb{F}_p$. Dimensions are ranks of evaluation matrices. Subspace claims are established by two-way containment, never by matching dimensions: two different 12-dimensional subspaces would otherwise be indistinguishable, and at degree ten there really are three distinct ones.

Modular arithmetic bounds claims in one direction only, and we use it only in that direction. A rank over $\mathbb{F}_p$ is an unconditional lower bound on the rank in characteristic zero whenever the matrix is the reduction of an integer matrix, which holds here because the coordinate basis is integral. Where an upper bound is needed it comes from counting an explicit spanning set, not from a modular computation: a space spanned by exactly k invariants has dimension at most k for free, and when its modular rank is also k its dimension is k over $\mathbb{Q}$ with no prime excluded. That covers A_{10}, P_{10}, G_{10} and B_{10}. It does not cover D_{10}, which is assembled by admitting directions that raise the rank modulo p; there 11 is a lower bound over $\mathbb{Q}$, so $\dim Q_{10} \leq 3$. A bad prime could only make the flow reach more than we report, never less.

The same principle settles the functional count. Replacing the finite-difference Jacobian by an exact analytic one — taken by amputation, with no step size and

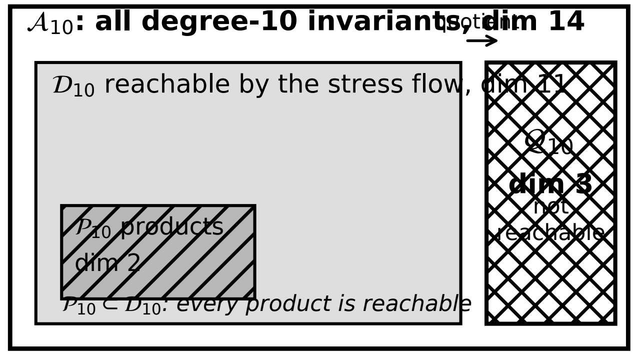

FIG. 1. The degree-ten invariant space and its subspaces, with exact dimensions. Products of lower-degree invariants lie inside the reachable sector, so the 3-dimensional quotient is not generated by anything of lower order. Shading is redundant with the labels; the figure reads in grayscale.

no rank tolerance, and validated by Euler homogeneity — gives rank 81 over the complete selection of 83 candidates, with 0 evaluation errors and 0 zero rows. An explicit 81×81 minor of the integral Jacobian has non-vanishing determinant, so $\mathrm{rank}_{\mathbb{Q}} \geq 81$ unconditionally. This meets the analytic count 81 from below; the upper half remains analytic, and the computation is not claimed to establish the generic rank by itself. That 83 candidates have rank 81 also says, plainly, that the candidates carry functional dependencies.

The spinor description is connected to the tensor description by the map $S_{ab} = \frac{1}{5!}F_{\mu_1\ldots\mu_5}(\Gamma^{\mu_1\cdots\mu_5})_{ab}$, built with every convention pinned and verified: its kernel is exactly the anti-self-dual 126 and its image exactly the gamma-traceless 126, both by span equality, with a left inverse that reproduces the self-dual projector exactly. Equivariance is verified under products of Clifford reflections, which generate the full rotation group.

One point about the frames is easy to get wrong and we state it explicitly. The oscillator realisation used on the spinor side has real signature $(5, 5)$, not Euclidean: a null frame cannot be Euclidean, since Euclidean signature has no isotropic vectors. Split and Lorentzian signatures both give $\star^2 = +1$ on five-forms, so real self-dual five-forms exist in each; but $(5, 5)$ and $(1, 9)$ are inequivalent real forms and the transition between them is not real. Over $\mathbb{F}_p$, where only the discriminant survives, they are congruent and the transition is exact — which is where the component-level comparison is done.

Physical consequence. The practical content is a statement about what a deformation scheme must contain.

Fix any construction whose degree-ten output lies in $\mathcal{D}_{10}$ — in particular any flow generated by the stress-tensor invariants of the seed theory, iterated to any order, and any composition of lower-degree results. Equa-

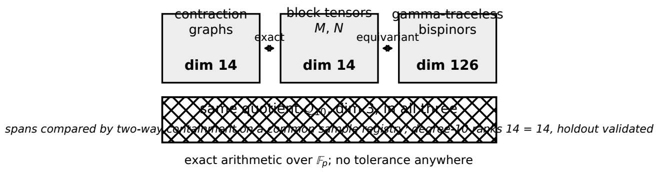

FIG. 2. The same quotient in three descriptions. Arrows are verified maps; numbers are ranks established by two-way span containment on a common sample registry, in exact arithmetic at several primes.

tion (1) says such a construction misses a 3-dimensional space of local interactions, and the End Matter exhibits an explicit degree-ten invariant with nonzero class in $\mathcal{Q}_{10}$: no choice of flow parameters produces it, because its image in the quotient is nonzero while every element of $\mathcal{D}_{10}$ has zero image.

The consequence for model building is concrete. A deformation scheme intended to cover ten-dimensional chiral four-form dynamics at this order must enlarge its generator set by at least three independent structures beyond the stress-tensor sector, and three suffice at this order. That is a requirement on the formulation, not a preference: it is forced by the dimension of the quotient.

We are careful about what this does *not* say. It is an order-by-order statement at degree ten, not an all-orders theorem. It constrains the interaction space, not the dynamics of any particular theory, and it makes no statement about supersymmetric completion, causality, or type IIB coefficients.

Discussion. The degree-ten interaction space of the ten-dimensional self-dual five-form is now known exactly, together with the part of it that stress-tensor flows reach. The gap is three-dimensional, explicit, and irreducible in the sense that it contains no products of lower-degree invariants.

What changes conceptually is the status of the flow construction in ten dimensions. In lower dimensions it is a complete parametrisation. Here it is a proper subspace of known codimension, and the complement is small enough to write down. The natural next questions are whether the codimension grows with order — degree twelve is not classified and we make no claim about it — and whether supersymmetry or causality selects among the three directions.

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[1] J. Hutomo, K. Lechner, and D. P. Sorokin, JHEP **02**, 147, arXiv:2509.14351.

[2] M. Cederwall, J. Hutomo, S. M. Kuzenko, K. Lechner, and D. P. Sorokin, J. Phys. A **59**, 065203 (2026), arXiv:2509.14350.

END MATTER

Hypotheses. The Theorem holds under the following, all of which are needed and none of which is implicit.

(H1) Locality and no derivatives. Invariants are Lorentz-invariant polynomials in the components of F alone. Terms containing ∂F are outside the class.

(H2) Self-duality. F is constrained to the 126-dimensional self-dual subspace of Λ^5, with $\star F = +F$ in signature $(-, +, \dots, +)$ and $\epsilon_{01\dots9} = +1$.

(H3) Fixed grading. Degree means polynomial degree in F. Degree ten is the graded piece, not a cumulative count.

(H4) The reachable sector. $\mathcal{D}_{10}$ is the span of degree-ten invariants generated by the stress-tensor flow of Ref. [1] acting on the free seed theory, closed under the products the flow generates. Its precise generator list is in the Supplemental Material. A different flow defines a different $\mathcal{D}_{10}$ and the theorem must be re-read accordingly.

(H5) No field redefinitions. Invariants are compared as polynomials, not modulo field redefinitions of F. Allowing redefinitions could only shrink the quotient and is a separate question.

(H6) Arithmetic. Dimensions are exact ranks over $\mathbb{F}_p$ at the primes listed in the Supplemental Material. Because the coordinate basis is integral, each is an unconditional lower bound on the characteristic-zero rank; the matching upper bounds come from explicit spanning sets, also exact.

The three representatives. Explicit compact representatives of $\mathcal{Q}_{10}$, in the pure-N presentation that avoids an index ambiguity present in the published expressions, are listed in Table I of the Supplemental Material together with their coordinates in the degree-ten basis and their images in the quotient. Removal minimality is checked directly: deleting any one representative drops $\dim \mathcal{Q}_{10}$ from 3 to 2.

What is computer-assisted, and how to check it. The dimensions in Eq. (1) are computer-assisted. They are not opaque: each is the rank of an explicitly recorded integer matrix, and the repository ships the matrices, the pivot rows and columns, and a script that recomputes each rank from the stored data without re-running any search. A reader who distrusts the enumeration can still check the ranks.

The enumeration itself — that the recorded candidates exhaust degree ten in the declared class — is the step that rests on the search. It is corroborated by an independent formula-free reverse recovery that reaches the same quotient span, and by the spinor-side computation, but it is not a symbolic proof and we do not present it as one.

What is not claimed. The reverse search is bounded and is not an exhaustive enumeration of all contraction sectors. No generating set for the full invariant ring is presented. Degree twelve is not classified. The three representatives are a certified choice, not canonical objects. Agreement at several primes is not a characteristic-zero identity, and no statement here relies on it being one.